Adaptive Traffic Signal Control Using Demand Forecasting and Rolling-Horizon Optimization

Younggun Kim[1] and Manuel A. Martinez Garcia[1]

Abstract— **Traffic signal control plays an important role in managing congestion and reducing vehicle delays at urban intersections. Traditional fixed-time signal controllers rely on predetermined timing plans that may not adequately respond to fluctuations in traffic demand. While adaptive controllers can respond to observed vehicle queues, they do not necessarily incorporate recurring temporal patterns or uncertainty in future arrivals. This project proposes a predictive traffic signal control framework that integrates Transformer-based traffic demand forecasting with rolling-horizon optimization. The forecasting model learns temporal demand patterns from historical vehicle arrivals and time-of-day information to predict traffic demand over a finite future horizon. These predictions are incorporated into a rolling-horizon optimization component that dynamically determines signal phases and green time allocations to minimize cumulative vehicle delay subject to operational constraints. Using an isolated-intersection simulation, we will compare the proposed controller against fixed-time and queue-responsive control under balanced demand, directional demand shifts, heavy congestion, and imperfect detector observations. We will evaluate whether incorporating forecasts improves traffic signal performance and investigate how forecasting errors affect control decisions.**

I. INTRODUCTION

Traffic congestion at signalized intersections remains a significant challenge in urban transportation systems, often contributing to increased travel times, vehicle delays, and inefficient utilization of roadway capacity. Conventional fixed-time traffic signal controllers allocate green time according to predetermined schedules, typically designed around representative traffic demand conditions. Although these controllers are relatively simple to implement, their performance may deteriorate when actual traffic conditions deviate from the assumptions used to develop their timing plans. Adaptive traffic signal control provides an alternative by adjusting signal timings in response to observed traffic conditions, potentially improving traffic flow and reducing delays.

One important characteristic of traffic demand is its temporal variability. Traffic arrivals often exhibit recurring patterns associated with time of day and day of the week. Recent research by ElKhouly et al. [2] demonstrates the potential of forecasting methods to capture multiple demand patterns in intersection traffic volumes. Additionally, Li and Savla [3] investigate adaptive model predictive control for traffic signal timing under unknown traffic demand and system parameters. More recently, Atef and Karam [1] combined traffic forecasting with constrained model predictive control to address recurring changes in demand.

These studies highlight the potential of integrating predictive information with optimization-based traffic signal control. However, an important question remains regarding how much demand forecasting improves actual control performance, particularly when observed traffic conditions deviate from historical patterns or measurements become unreliable.

In this project, as illustrated in Figure 1, we propose developing an adaptive traffic signal controller that combines demand forecasting with rolling-horizon optimization. Our approach will estimate short-term traffic arrivals using recurring demand patterns and continuously update these predictions using available detector observations. The resulting forecasts will inform an optimization model that determines feasible green time allocations while considering vehicle queues, discharge capacities, minimum and maximum green times, and phase-transition requirements.

We will implement the proposed framework using the provided isolated-intersection simulation

[1]Department of Civil and Environmental Engineering, University of California, Los Angeles (UCLA), Los Angeles, California, USA.

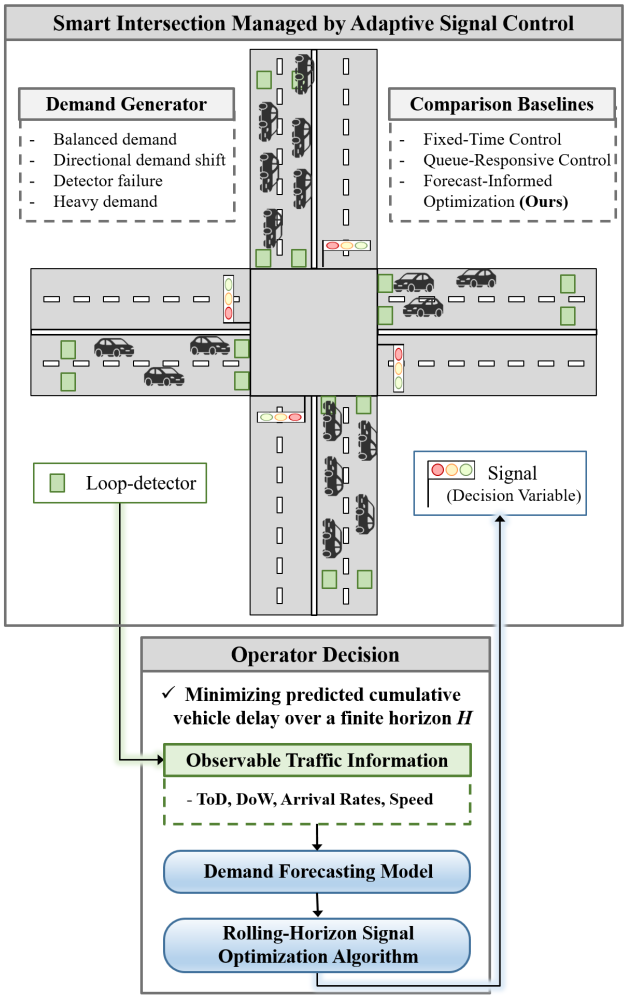


Fig. 1. Overview of the proposed adaptive traffic signal control framework integrating Transformer-based demand forecasting and rolling-horizon optimization in an isolated intersection.

scaffold and compare its performance against justified fixed-time and queue-responsive controllers. Through simulation experiments involving changing traffic demand, heavy congestion, and detector failures, we aim to investigate the relationship between forecasting accuracy and traffic signal performance, with particular emphasis on total intersection delay, throughput, queue lengths, and service imbalance across approaches.

II. PROPOSED METHODOLOGY

Our proposed framework consists of three main components: demand traffic forecasting, rolling-horizon signal optimization, and simulation-based performance evaluation.

A. Intersection Model

We consider an isolated four-way signalized intersection consisting of northbound, southbound, eastbound, and westbound approaches.

For simplicity, the initial model will consider two compatible traffic signal phases:

- Phase 1: Northbound and southbound through movements.
- Phase 2: Eastbound and westbound through movements.

Vehicle arrivals, queue formation, and stop-line discharge will be represented using the provided simulation scaffold. The initial scope excludes pedestrian movements and turning movements.

Signal operation will satisfy minimum and maximum green durations, yellow and all-red clearance intervals, movement compatibility, and discharge capacity constraints.

B. Transformer-Based Traffic Demand Forecasting

We propose a Transformer-based forecasting model to predict short-term vehicle arrival rates using historical traffic observations and temporal information.

Let $\mathcal{A}$ denote the set of intersection approaches, and let $\boldsymbol{\lambda}_t$ denote the vector of observed arrival rates at time t.

At each prediction time t, the model takes a historical sequence of length L as input:

$$\mathbf{X}_t = [\boldsymbol{\lambda}_{t-L+1}, \ldots, \boldsymbol{\lambda}_t]. \tag{1}$$

The historical demand sequence is augmented with time-of-day (ToD) and day-of-week (DoW) features to capture recurring temporal patterns.

The Transformer, parameterized by θ, predicts the vehicle arrival rate for each approach i at the next time step:

$$\hat{\lambda}_{i,t+1} = f_\theta(\mathbf{X}_t, \mathbf{C}_t)_i, \qquad i \in \mathcal{A}, \tag{2}$$

where $\mathbf{C}_t$ contains the corresponding ToD and DoW features.

To obtain forecasts over a future horizon H, the predicted arrival rates are recursively fed back into the historical input sequence. At each iteration, the oldest observation is removed and the latest prediction is appended along with its corresponding

temporal features. This process is repeated H times using the same Transformer model.

The resulting demand forecasts will be incorporated into the rolling-horizon signal optimization model to determine future signal control decisions.

C. Rolling-Horizon Signal Optimization

The estimated arrival rates will be incorporated into a rolling-horizon optimization framework that determines future signal phases and green time allocations.

Let $q_{i,t}$ represent the queue length at approach i and time t. The queue dynamics may be represented by

$$q_{i,t+1} = q_{i,t} + a_{i,t} - d_{i,t}, \tag{3}$$

where $a_{i,t}$ denotes accepted vehicle arrivals and $d_{i,t}$ denotes the number of discharged vehicles.

For finite-storage approaches, arrivals that cannot enter the intersection model will be recorded as unmet demand rather than discarded.

The optimization objective will seek to minimize predicted cumulative vehicle delay over a finite horizon H.

A preliminary objective can be expressed as

$$\min_{\mathbf{u}} \mathbb{E}\left[\sum_{\tau=t}^{t+H-1} \sum_{i\in\mathcal{A}} q_{i,\tau}\Delta t\right], \tag{4}$$

where $\mathbf{u}$ represents feasible signal control decisions and Δt is the simulation time step.

The cumulative queue term serves as a measure of total vehicle waiting time. Additional penalties may be introduced to discourage excessive service imbalance between approaches.

The optimization will enforce signal phase compatibility, minimum and maximum green times, clearance intervals, and vehicle discharge constraints.

At each decision interval, the controller will implement the first feasible control decision, update its observations, and repeat the optimization process.

III. SIMULATION AND EVALUATION

A. Controller Comparison

We propose evaluating three traffic signal control strategies:

1) Fixed-Time Control: A predetermined signal timing plan calibrated to representative traffic demand.
2) Queue-Responsive Control: A controller that adjusts signal decisions based on observed vehicle queues.
3) Forecast-Informed Optimization: A rolling-horizon controller that incorporates demand forecasts and updated traffic observations.

All controllers will operate under the same signal phase and clearance constraints to ensure a fair comparison.

B. Simulation Scenarios

We will evaluate controller performance under four traffic conditions:

1) Balanced demand: Comparable arrival rates across intersection approaches.
2) Directional demand shift: A sudden increase in vehicle arrivals along one intersection direction.
3) Detector failure: Missing, inaccurate, or delayed observations of traffic conditions.
4) Heavy demand: Arrival rates approaching or exceeding available intersection service capacity.

When observations become unreliable, the proposed controller will apply a predefined fallback timing plan.

All experiments will use identical demand realizations, initial queues, and simulation horizons across controllers.

C. Performance Metrics

Controller performance will be evaluated using:

- Total and approach-specific vehicle delay.
- Intersection throughput.
- Maximum and residual queue lengths.
- Unmet vehicle demand.
- Service imbalance between approaches.

We will also examine signal phase timelines and the corresponding queue evolution to verify the controller's behavior and signal transition constraints.

D. Statistical Analysis

Repeated simulation experiments will be conducted using different random seeds to capture variability in vehicle arrivals.

We will compare controller performance using paired experimental observations, summary statistics, and confidence intervals.

Additionally, we propose conducting a sensitivity study to investigate how forecast accuracy and observation delays affect optimization performance.

An ablation experiment may compare the same optimization controller with and without seasonal forecasting to isolate the contribution of the statistical model.

IV. EXPECTED CONTRIBUTIONS

The primary objective of this project is to investigate the benefits and limitations of incorporating demand traffic forecasting into adaptive traffic signal control.

We aim to evaluate whether the predicted traffic information improves short-term signal decisions relative to conventional fixed-time and queue-responsive controllers.

Additionally, we seek to understand how forecasting errors and unreliable observations influence optimization performance under changing traffic conditions.

The results may help identify circumstances in which forecast-informed optimization provides measurable benefits and circumstances in which simpler traffic signal control strategies remain competitive.

If successful, this work could demonstrate the potential of forecast-informed traffic signal control at intersections where sufficient historical and real-time traffic data are available, potentially reducing traffic delays and improving overall intersection efficiency.

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